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Experience

Sep 2026 – Present
Bénin

Research Engineer

Ai4Innov Technologies

Supervisor: Dr. [Gilles Hacheme](#)

- Geospatial AI & Satellite Imagery Fellowship: working on machine learning pipelines that turn satellite imagery into insights for climate resilience and food security.

Jul 2026
Bénin

Applied Machine Learning Mentor

Sèmè City

- Mentored the top 20 participants of the National Olympiad of Artificial Intelligence (NOAI) 2026 in machine learning and technical problem-solving.
- At the International Olympiad in Artificial Intelligence (IOAI), two mentees earned a bronze medal and an honorable mention, respectively, and the team ranked third in Africa. [\[Link\]](#)

Mar 2026 – Aug 2026
Bénin

Research Intern

LRSIA, University of Abomey-Calavi

Supervisor: Dr. [Ratheil Houndji](#)

- Designed a MILP model for the Multi-Product Vehicle Routing Problem with Changeover Costs (MPVRP-CC).
- Bachelor thesis: proposed a hybrid Constraint Programming and construction heuristic approach for the MPVRP-CC, using sequence variables, a recent CP paradigm. [\[Documentation\]](#)
- Supervised the second-year AI class project (Machine Learning course) and contributed to the [ifri-mini-ml-lib](#) pedagogical library.
- Contributed to the lab's internal projects and to technology monitoring.
- **Tools:** Gurobi, MaxiCP, MiniCP, Python, Java

Jun 2025 – Jan 2026
Bénin

AI Developer

IFRI & Tekbot Robotics

Supervisor: [David Akpovi](#)

- Developed and deployed an AI vision model and robotic controls for an automated waste-sorting system, allowing a robot arm to recognize and sort trash in real time to improve urban recycling.
- **Tools:** CNN, YOLOv5, PyTorch, Jetson Nano, CUDA, ONNX, TensorRT

Jun 2025 – Aug 2025
South Africa

Machine Learning Engineer Intern

eTihuku

Supervisor: Dr. [Jeremiah Olamijuwon](#)

- Built Review Scraper, an asynchronous API for collecting and aggregating customer reviews from Google Maps, HelloPeter, and X (Twitter), with background task processing and a scalable architecture.
- Worked on SentiMaster, an AI platform that collects customer feedback from social media and phone calls, automatically translates and analyzes hidden sentiments, emotions and topics, and helps companies predict customer churn.
- **Tools:** FastAPI, Playwright, PyTorch, Transformers, BERT models, Airflow, Redis

Research & Presentations

Aug 2026
Lagos, Nigeria

Deep Learning Indaba

A MILP Framework for Multi-Product Vehicle Routing Problem with Changeover Costs [\[Poster\]](#)

Nov 2025
Abomey-Calavi, Bénin

Benin Workshop on AI

Evolutionary Algorithms: Principles, Variants, Application [\[Material\]](#)

Aug 2025
Abomey-Calavi, Bénin

ACP Summer School

Constraint Programming for Sustainable Development [\[Certificate\]](#)

Aug 2025
Ouidah, Bénin

Young Girls' Leadership Camp

Young Woman, Take the Lead in Shaping Your Future with AI [\[Link\]](#)

Co-presented with [Marie-Paule Lima](#)

Awards

2023 - 2026	Beninese Government Scholarship , awarded for academic excellence in the Baccalaureate .
2020 - 2026	Vallet Foundation Scholarship , awarded for academic excellence in the BEPC .
2025	2nd Place , Tekbot Robotics Challenge 2025. [Link]
2024	1st Prize , IndabaX Benin Republic Hackathon 2024. [Certificate]

Volunteering

Feb 2025 - Present	FRIARE Africa Participate in awareness initiatives on AI ethics, transparency and inclusion. [Link]
Jun 2025 - Aug 2026	THE LAB Co-led an AI awareness community in Bénin. [Link]
Jan 2025 - Jul 2026	IFRI AI & IoT Club Co-led the AI & IoT student club: delivered introductory sessions on Machine Learning and Optimization Algorithms, coordinated technical activities, and co-supervised AI projects. [GitHub]
Mar 2026 - Apr 2026	Applied Machine Learning Days (AMLD) 2026 Co-organized the first edition of the Applied Machine Learning Days in Benin Republic. [Certificate]

Education

Nov 2023 - Present Bénin	Bachelor of Science in Artificial Intelligence <i>IFRI, University of Abomey-Calavi</i> Thesis: <i>Solving the Multi-Product Vehicle Routing Problem with Changeover Costs: A Hybrid Heuristic-Constraint Programming Approach.</i> Expected completion: November 2026
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Selected Projects

- **le-foncier-intelligent**: Developed a geospatial land-analysis pipeline combining multimodal OCR, geocoding, and satellite imagery to extract and assess information from topographic documents. Processed geospatial data using GeoPandas, Shapely and pyproj to support automated land-use analysis. [\[Code Repository\]](#)
- **automated real estate mapping**: Designed an ETL pipeline to link address data, cadastral information, companies, and executives, in order to automatically generate enriched datasets for prospecting and analysis. [\[Code Repository\]](#)
- **MDVRP**: Designed and implemented optimization algorithms (ALNS, PSO, Simulated Annealing, Linear Programming) for multi-depot vehicle routing (MDVRP) under capacity and distance constraints. [\[Code Repository\]](#)
- **Opti'plan**: Co-built an automated scheduling model that reduced the university's thesis defense planning time from 3 days to under 5 minutes using Constraint programming.
- **insurance-recommendation-system**: Developed an end-to-end machine learning pipeline for insurance product recommendation, including data processing, model training and comparison (XGBoost, CatBoost, Deep Learning), with a Flask API and web interface supporting prediction and automated model retraining. [\[Code Repository\]](#)
- **ifri-mini-ml-lib**: Co-built an open-source Python library that reimplements classic AI algorithms from scratch for educational purposes; published on PyPI and used by IFRI students. [\[Code Repository\]](#)

Skills

Programming	Python, Java, R, Bash
Machine Learning	PyTorch, JAX, Scikit-learn, Transformers
Optimization	Linear and Constraint Programming, Metaheuristics; Gurobi, MaxiCP
Tools	Git, Linux, Docker, FastAPI, GitHub Actions
Languages	French (native), English (B2)

Research Interests

Combinatorial optimization, industrial planning and scheduling problems, applied machine learning for social good.